Samawel JABALLI, PhD

Currently Teaching Assistant, Univ. of Bordeaux (College of Science and Technology, Computer Science Unit) — Active Researcher, LaBRI

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LinkedIn — Google Scholar — DBLP — GitHub — ORCID — Acclaim

Summary — *Researcher in Natural Language Processing with a dual PhD from the Univ. of Bordeaux and the Univ. of Monastir. My research focuses on multimodal learning and social media analysis, with an emphasis on robustness and interpretability in AI systems. I have experience in both research and university-level teaching in computer science.*

Education

PhD in Computer Science, Doctoral School of Math & Comp. Sci. — Univ. of Bordeaux ***Feb. 2023–Jun. 2025***

- Extraction, Transcription, and Analysis of Multilingual Public Opinions from Social Media Networks.
- Host Laboratory : **LaBRI** - Laboratoire Bordelais de Recherche en Informatique (**CNRS**/ Univ. of Bordeaux).
- Co-tutelle Partner: Doctoral School of Information Science & Technology - Faculty of Sciences, Univ. of Monastir.

PhD in Computer science, Doctoral School of Information Sci. & Tech. — Univ. of Monastir ***Feb. 2022–Jun. 2025***

- Data Mining and Social Media Analysis: Applications related to Pandemic Governance.
- Host Laboratory : **RLANTIS** - Research Laboratory in Algebra, Numbers theory and Intelligent Systems.
- Co-tutelle Partner : Doctoral School of Mathematics & Computer Science, University of Bordeaux.

M.Sc. in Modeling of Automatic Reasoning Systems — Faculty of Sciences, Univ. of Monastir ***Sep. 2019–Nov. 2021***

- First-Class Honours, *Specialization in Natural Language Processing.*
- Courses: AI Fundamentals (Advanced Statistics, Optimization), Machine & Deep Learning, Natural Language Processing, Computer Vision, Knowledge Representation, Data Mining, Big Data ...

B.Sc. in Computer Science — Faculty of Sciences, Univ. of Monastir ***Sep. 2016–Nov. 2019***

- Highest Honours, *Curriculum focused on Computer Science fundamentals, from algorithms to systems architecture.*
- Courses: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Software Engineering, Computer Networks, Database Systems, Cryptography & Security...

Publications

PEER-REVIEWED PUBLICATIONS (CONFERENCE PROCEEDINGS & JOURNAL ARTICLES)

– **When Silence Matters: How VAD-driven Whisper Architectures reshape Temporal Alignment in Low-Resource Multilingual ASR** ***Sep 2026***
Published in Proc. 18th International Conference on Computational Collective Intelligence (ICCCI 2026).

– W. Tlili, **S. Jaballi**, et al., **Quantifying Clinical Risk in Generative Radiology Reporting: MCRS, A Medical Composite Reliability Score** ***Sep 2026***
Accepted in Proc. 32nd International Conference on Cooperative Information Systems (CoopIS 2026).

– M. J. Hazar, **S. Jaballi**, et al., **"A Hybrid E-Learning Recommendation System Incorporating User Reviews and Ratings for Enhanced Course Selection"** ***Sep 2026***
Published in Proc. 18th International Conference on Computational Collective Intelligence (ICCCI 2026).

– A. N. P. Dieng, **S. Jaballi**, et al., **Toward a Unified Multimodal Framework for Chest X-Ray Analysis: A Hybrid Convolutional–State Space Approach** ***Jul. 2026***
Published in Proc. 12th International Conference on Control, Decision and Information Technologies (CoDIT 2026).

– A. Mahjoubi, **S. Jaballi**, et al., **TUNIPROD: An Optimized ProdLDA-Based Multi-Stage Framework for Cross-Lingual Topic Modeling of Code-Switched Public Discourse** ***Jul. 2026***
Published in Proc. 12th International Conference on Control, Decision and Information Technologies (CoDIT 2026).

– **ResNet-Based Pandemic Keyword Spotting in Continuous Multilingual Speech: A Study in UNESCO's Audio Messages for Rapid Health Response** ***Nov. 2025***
Published in Proc. 17th Int. Conf. on Computational Collective Intelligence (ICCCI), Springer (CCIS series).

– **Exploring Unsupervised Word Representation Models and Neural Networks for Informal Multilingual Text against Covid-19 Social Media Content** ***Apr. 2024***
Published in Proc. 16th Int. Conf. on Asian Intelligent Information and Database Systems (ACIIDS), Springer (LNAI series).

– **Decoding Multilingual Topic Dynamics and Trend Identification through ARIMA Time Series Analysis on Social Networks: A Novel Data Translation Framework Enhanced by LDA/HDP Models** ***Jan. 2024***
Published in Journal of Electrical and Computer Engineering, vol. 2024.

– **Deep Bidirectional LSTM Network Learning-Based Sentiment Analysis for Tunisian Dialectical Facebook Content During the Spread of the Coronavirus Pandemics** ***Sep. 2023***
Published in Proc. 15th Int. Conf. on Computational Collective Intelligence (ICCCI), Springer (CCIS series).

– **Sentiment Analysis of Tunisian Users on Social Networks: Overcoming the Challenge of Multilingual Comments in the Tunisian Dialect** ***Sep. 2022***
Published in Proc. 14th Int. Conf. on Computational Collective Intelligence (ICCCI), Springer (LNAI Series).

PREPRINTS & ACTIVE SUBMISSIONS (UNDER REVIEW / IN PREPARATION)

- A. N. P. Dieng, **S. Jaballi**, et al., **MAF-Net: A Unified Hybrid Convolutional–SSM Framework for Multimodal Chest X-ray Triage, Segmentation, and Automated Reporting** ***Apr. 2026***
  Submitted to IEEE Transactions on Image Processing.
- A. N. P. Dieng, **S. Jaballi**, et al., **Breaking the Multimodal Shortcut: A Triple-Memory Decoder with Constrained Gating for Radiology Report Generation** ***Apr. 2026***
  Submitted to Expert Systems with Applications.
- **Unsupervised Approach for Augmenting Low-Resource UNESCO Multilingual COVID-19 Audio Dataset for Speech Recognition Tasks (from ∼30 Minutes to 6 Hours)** ***Jun. 2025***
  Under review at Computer Speech & Language.
- **Analyzing Multilingual Conversations During COVID-19: An Imbalanced Class-Ensemble Learning Approach with Reweighted AdaBoost-SVM for Code-Switched Text Classification** ***Feb. 2024***
  Under review at Multimedia Tools and Applications.

Research Experience

Postdoctoral Researcher (ATER) — Laboratoire Bordelais de Recherche en Informatique (LaBRI) ***Sep. 2025 – Present***

- **Dialect Normalization:** *Developing adaptive neural models to process informal and dialectal content.*
  Keywords: *Meta-learning, zero-shot/few-shot transfer, adaptive language modeling, self-regularizing decoders.*
- **Multimodal Self-Supervised Analysis:** *Leveraging foundation models for multilingual text and audio to build universal representations for large-scale informal content analysis.*
  Keywords: *Foundation models, self-supervised learning, cross-modal alignment, speech-text fusion, cross-lingual transfer.*
- **Sentiment Analysis with Enriched Signals:** *Integrating dynamic knowledge graphs and prosodic features into opinion analysis pipelines, using GNNs to model discourse dynamics in polarized social contexts.*
  Keywords: *GNNs, dynamic knowledge graphs, prosodic features, multimodal sentiment analysis, discourse-aware modeling.*

Invited Researcher — NTIS Research Centre, University of West Bohemia, Pilsen, Czech Republic ***Jul. – Aug. 2025***

- *Invited by **Pavel Kral** to define the scope for a survey on Cross-lingual Aspect-Based Sentiment Analysis and to design a proof-of-concept for adapting a Czech transformer-based model to informal French.*

Doctoral Researcher — LaBRI (Univ. of Bordeaux) with RLANTIS (Univ. of Monastir) ***Feb. 2022 – Jun. 2025***
*PhD supervised by **Nicolas Henri** and **Mounir Zrigui**.*

- **Laboratoire Bordelais de Recherche en Informatique (LaBRI), Univ. of Bordeaux** *Feb. 2023 – Jun. 2025*
  - **Topic Trend Extraction:** *Designed probabilistic and time-series models (e.g., LDA, HDP, ARIMA) to uncover latent themes and model the evolution of societal concerns.*
  - **Multilingual Speech Processing:** *Built a hybrid system for automatic transcription and keyword detection, leveraging optimized architectures (FasterWhisper, WhisperX) and voice activity detection models (Silero VAD, Pyannote VAD).*
  - **Acoustic & Contextual Representation Learning:** *Combined deep neural networks (ResNet) with acoustic (MFCC) and contextual (M-BERT) embeddings to enhance the identification of critical information in continuous speech.*
- **Research Lab. in Algebra, Numbers theory & Intelligent Systems, Univ. of Monastir** *Feb. 2022 – Jan. 2023*
  - **Code-Switched Sentiment Analysis:** *Developed a hybrid architecture (Bi-LSTM + AdaBoost-SVM) with adaptive cost-sensitive learning to improve opinion classification in multilingual and dialectal contexts.*

Data Engineer Intern — Statistics & Optimization Team, Institut de mathématiques de Toulouse ***May – Oct. 2022***

- *Architected and deployed a scalable data pipeline to construct a novel Linguistic-Quantitative Corpus, synthesizing over 8,000 French press articles with longitudinal public health data (Supervised by **Frederic Ferraty**).*

Master's Thesis Researcher — Research Lab. in Algebra, Numbers theory & Intelligent Systems ***May – Oct. 2021***

- **Novel Pipeline for Low-Resource Sentiment Analysis:** *Developed a Bi-LSTM-based pipeline for Tunisian dialect sentiment analysis, integrating custom preprocessing and a multilingual benchmark corpus to improve code-switched modeling (Supervised by **Mounir Zrigui**).*

Teaching and Academic Experience

TEACHING EXPERIENCE

Temporary Teaching and Research Fellow (ATER: Full-time) — Univ. of Bordeaux ***Sep. 2025 – Present***

- **S1 Responsibilities:** *Leading lectures and practical lab sessions for the following core undergraduate modules:*
  - Elementary Data Structures & Algorithms (**4TIN302U**).
  - Fundamentals of Algorithms and Programming (**4TTV109U**).
  - Computer Science 1 (Introduction to Python for Majors) (**4TPU146U**).
- **S2 Responsibilities:** *Scheduled to teach advanced courses and supervise graduate-level projects:*
  - Advanced Data Structures (Trees & Graphs in OCaml) (**4TIN402U**).
  - Master's Programming Project Supervision (**4TIN802U**).
  - Compilation (machine-oriented TD sessions with integrated lectures) (**4TIN603U**).

University Lecturer (Part-Time) — Private Polytechnic School of Monastir ***Jan. – Jun. 2024***

- **S2 Responsibilities:** *Taught specialized modules (lectures & labs) to undergraduate Software Engineering majors in:*
  - Data Warehousing (Evening Course for 5th-Year Students).
  - Machine Learning (Day Course for 4th-Year Students).

University Lecturer (Part-Time) — Faculty of Sciences, Univ. of Monastir ***Sep. – Jun. 2023***

- **S1 Responsibilities:** *Instructed foundational undergraduate lab sessions in:*
  - Advanced Python Programming.
  - Relational Databases.
- **S2 Responsibilities:** *Delivered the advanced undergraduate (L3) module, including lectures and practical sessions, for:*
  - Data Warehousing (SQL + Python).

SUPERVISION & MENTORSHIP

Co-Supervision of MSc Thesis (Hybrid) — RLANTIS/ Faculty of Sciences, Univ. of Monastir ***Jan. 2026 – Present***
Project: Multimodal Architectures for Computer-Aided Radiology Report Generation — with Mounir Zrigui

- *Supported the methodological design and experimental development of multimodal models for chest X-ray report generation, contributing to research on Triple-Memory decoding, constrained gating, and visual-text alignment.*

Co-Supervision of MSc Thesis (On-site) — RLANTIS/ Faculty of Sciences, Univ. of Monastir ***Jan. 2026 – Present***
Project: Hallucination Detection and Mitigation in Multimodal Medical AI Systems — with Mounir Zrigui

- *Studied hallucinations in multimodal medical AI; developed a vision–language architecture for detection and mitigation.*

Co-Supervision of MSc Thesis (Remote) — Ravensbourne University London ***Sep. 2025 – Jan. 2026***
Project: Enhancing Embedding Representations in NLP Models for Informal Digital French Content — with Ricky Lim

- *Co-architected the research framework for adapting French language models to informal text, defining an end-to-end methodology encompassing the construction of a noisy French corpus, development of embedding-space diagnostic protocols, and exploration of domain-specific fine-tuning and tokenizer vocabulary extension.*

Supervision of B.Sc. Thesis (On-site) — RLANTIS/ Faculty of Sciences, Univ. of Monastir ***Jan. – Jun. 2024***
Project: Whisper-Based Neural Model for Multilingual ASR in COVID-19 Scenarios: Benchmarking on UNESCO Podcasts

- *Defined the project's methodology, including literature review, data augmentation, MFCC feature extraction, model selection, fine-tuning, and comparative benchmarking of transformer models. The adjusted Whisper-based model achieved a 9% WER, outperformed baselines, and produced a functional multilingual transcription application using UNESCO COVID-19 podcasts.*

Supervision of B.Sc. Thesis (Hybrid) — TotalEnergies/ Faculty of Sciences, Univ. of Monastir ***Jan. – Jun. 2024***
Project: Development of a Web Component for a Maritime Logistics Module in an Enterprise Trading Platform (©MetaFactory)

- *Mentored the student on the design and development of a mission-critical web component for managing routing and tracking of oil-carrying vessels within TotalEnergies' enterprise trading platform, guiding full-stack development (ReactJS, HTML, CSS for front-end; .NET Core + C for back-end), modular design, API integration, and deployment in enterprise environments.*

Co-Supervision of MSc Thesis (On-site) — RLANTIS/ Faculty of Sciences, Univ. of Monastir ***Jan. – Jun. 2023***
Project: Topic Modeling and Trend Analysis on Informal, Code-Switched Social Media Exchanges — with Mounir Zrigui

- *Co-architected the research framework and guided the implementation of an NLP pipeline analyzing 160K informal, code-switched social media texts, including corpus aggregation, hybrid translation to English (dictionaries, crowd-sourcing, GPT-3), topic modeling (LDA, HDP), and trend analysis via ARIMA, producing validated insights on social trends* (**DOI & Paper** 🔗).

Honors & Awards

- **Best Student Paper Award Nomination, ICCCI 2025— Ho Chi Minh City, Vietnam** ***Sep. 2025***
  Paper (🔗) acknowledged by the IPC and selected for its originality and applied contribution to public health communication.
- **Postdoctoral Travel Grant, LaBRI — Ho Chi Minh City, Vietnam** ***Jun. 2025***
  Merit-based award to support the presentation of my peer-reviewed paper(🔗) at the 17th ICCCI, Ho Chi Minh City, Vietnam.
- **Second Prize Winner, PEPITE ECA Doctoral-Enterprise Challenge — OpenCampusInnov, La Rochelle** ***Dec. 2024***
  Awarded for the "Print My Boat" project, a joint team of four PhD candidates from the Univ. of Bordeaux, La Rochelle, & Pau.
- **Semi-finalist, Bordeaux Population Health Young Researchers' Scientific Day — Univ. of Bordeaux** ***Oct. 2024***
  Semi-finalist in a competitive doctoral public health forum, advancing through 5 thematic Q&A round tables.
- **4thPlace, AFCP Doctoral Showcase — Association Francophone de la Communication Parlée, Avignon** ***Apr. 2024***
  Awarded 4th pl. and "Audience Favorite" award for a 10-min presentation of doctoral research to a non-specialist audience.
- **Alternance Award for PhD Studies in Computer Science (Co-tutelle, Univ. of Bordeaux)** ***Fev. 2023***
  Awarded by the Univ. of Monastir, providing full social and financial coverage 6 months/year for the entire doctoral program.
- **Prize for the Best University Average in M.Sc. in Computer Science — Faculty of Sciences of Monastir** ***Dec. 2021***
  Awarded for graduating with honors and ranking 1st in the M.Sc. program.

Conference Presentations & Invited Talks

Conferences Presentations

- **When Silence Matters: How VAD-driven Whisper Architectures reshape Temporal Alignment in Low-Resource Multilingual ASR** — *Pres. at 18th ICCCI*, **Heraklion, Greece.** ***Sep. 2026***
- **A Hybrid E-Learning Recommendation System Incorporating User Reviews and Ratings for Enhanced Course Selection** — *Pres. at 18th ICCCI*, **Heraklion, Greece.** ***Sep. 2026***
- **Toward a Unified Multimodal Framework for Chest X-Ray Analysis: A Hybrid Convolutional–State Space Approach** — *Pres. at 12th CoDIT*, **Bari, Italy.** ***Jul. 2026***
- **TUNIPROD: An Optimized ProdLDA-Based Multi-Stage Framework for Cross-Lingual Topic Modeling of Code-Switched Public Discourse** — *Pres. at 12th CoDIT*, **Bari, Italy.** ***Jul. 2026***
- **ResNet-Based Pandemic Keyword Spotting in Continuous Multilingual Speech: A Study in UNESCO's Audio Messages for Rapid Health Response** — *Pres. at 17th ICCCI*, **Ho Chi Minh City, Vietnam.** ***Nov. 2025***
- **Exploring Unsupervised Word Representations Models and Neural Networks for Informal Multilingual Text Against Covid-19 Social Media Content** — *Pres. at 16th ACIIDS*, **Ras Al Khaimah, UAE.** ***Apr. 2024***
- **Deep Bidirectional LSTM Network Learning-Based Sentiment Analysis for Tunisian Dialectical Fb Content During the Spread of the Coronavirus Pandemic** — *Pres. at 15th ICCCI*, **Budapest, Hungary.** ***Sep. 2023***
- **Sentiment Analysis of Tunisian Users on Social Networks: Overcoming the Challenge of Multilingual Comments in the Tunisian Dialect** — *Pres. at 14th ICCCI*, **Hammamet, Tunisia.** ***Sep. 2022***

Invited Talks, Seminars & Workshops

- **Invited Talk, PEPITE France Annual Student-Entrepreneurship Day (PEIPS) — Univ. Paris-Saclay** ***May. 2026***
  Joined the organizing committee as a past prize-winner to share experience on bridging PhD research with industry.
- **Invited Presentation, Thematic Day on Low-Resource NLP, — GDR/TAL (CNRS), Grenoble, France** ***Dec. 2025***
  Invited by the CNRS (GDR/TAL) network to present my postdoctoral research on Low-Resource Dialect Normalization.
- **Invited Speaker, AFoDIB Doctoral Seminar (Sémi-doc) — LaBRI, Bordeaux, France** ***Apr. 2024***
  Presented a state-of-the-art survey on NLP approaches for informal and code-switched French text.
- **Workshop Delivered, Mini Symposium : Brain & Language Models — Inria Center, Univ of Bordeaux** ***Jan. 2024***
  Conducted a practical training session on Cross-lingual Aspect-Based Sentiment Analysis for multilingual datasets.
- **Invited Presentation, Annual Doctoral Day — EDSTI/ Faculty of Sciences, Univ. of Monastir** ***Jun. 2023***
  Presented my first-year doctoral research findings on Code-Switched Sentiment Analysis to faculty and peers.

Community Engagement & Scientific Affiliations

- **Track Chair, Trustworthy Foundation Models and Large Language Models for Low-Resource, Multimodal and Informal NLP (TRUST4NLP) — 19th Asian Conf. on Intelligent Information and Database Systems (ACIIDS), Bali, Indonesia** ***Apr. 2027***
- **PC Member, Special Session on Advances in Low-Resource Multimodal, Speech, and Language Processing (LR-MSLP) — 19th Asian Conf. on Intelligent Information and Database Systems (ACIIDS), Bali, Indonesia** ***Apr. 2027***
- **TPC Member — 18th Int. Conf. on Management of Digital Ecosystems, A Coruña, Spain** ***Sep. 2026***
- **PC Member, Special Session on Speech and Text Processing for Low-Resource Languages (STP-LRL) — 18th International Conference on Agents and Artificial Intelligence (ICAART), Marbella, Spain** ***Mar. 2026***
- **TPC Member — IEEE International Conference on Consumer Electronics – Taiwan (ICCE-Taiwan)** ***Feb. 2026***
- **Membership Associate** — Société Informatique de France ***Sep. 2025***
- **Participant, 58thAnnual Meeting of the Societas Linguistica Europaea — Univ. Bordeaux Montaigne** ***Aug. 2025***
- **Reviewer — Journal of Data Science and Intelligent Systems** ***Jul. 2025 – Present***
- **Participant, "Pie & AI" Special Meetup hosted by DeepLearning.AI — Lisbon, Portugal** ***Apr. 2025***
- **Senior PC Member, Special Session on Artificial Intelligence and Social Dynamics in Online Communities (AISOC) — 17th International Conference on Computational Collective Intelligence (ICCCI), Ho Chi Minh City, Vietnam** ***Nov. 2025***
- **Membership Associate** — Sigma Xi, The Scientific Research Honor Society ***Feb. 2024***
- **TPC Member — 6th Int. Conf. on Soft Computing and its Engineering Applications (icSoftComp)** ***Dec. 2024***
- **Editorial board member — Indonesian Journal of Electrical Engineering and Informatics** ***Sep. 2024***
- **Reviewer — Neural Computing and Applications** ***Jul. 2024 – Present***
- **Reviewer — Journal of Ambient Intelligence and Humanized Computing** ***Apr. 2024 – Present***
- **PC Member, Special Session on ML Approaches and Smart Systems for Content Analysis (MLSS-CA) — 16th Asian Conference on Intelligent Information and Database Systems (ACIIDS)** ***Apr. 2024***
- **Organizing Committee Member, Doctoral Symposium on Arabic NLP — RLANTIS/ Monastir, Tunisia** ***May. 2023***

Skills Summary

- **Programming Languages:** *Python (advanced), C/C++, Java, OCaml.*
- **Databases:** *SQL, MySQL, PostgreSQL, PL/SQL, NoSQL systems; data modeling, querying, and large-scale data handling.*
- **Design Tools & Agile Methods:** *Adobe Photoshop, UML, Scrum, Git.*
- **Languages:** *Arabic (native), French (fluent), English (professional working proficiency).*
- **AI & Data Science Specializations:** *Prompt Engineering Specialization (Vanderbilt University, 2024); IBM Data Science Professional Certificate (IBM, 2024); IBM AI Engineering Professional Certificate (IBM, 2024);*
- **Others:** *Course design using Moodle and Latex, Experience with AI-assisted development tools (GitHub Copilot, Claude).*